Panskura Banamali College (Autonomous)

B.Sc.(H) 6th Semester Examination -2022 (Online Mode)

Subject: Computer Science **Paper: DSE4T**

Paper Name: Soft Computing

F.M.: 40 **Time : 2 Hours**

Answer any four questions : **4x10=40**

1. Explain support , scalar cardinality ,core ,height ,normal and sub-normal of a fuzzy set. 10
2. A) Using your own intuition and definition of universe of discourse, plot membership functions for liquid level in the tank. (empty, very less, less, full, very full).
   b) Using inference approach, find the membership values for each of the triangular shapes (I, R, IR, E, T) for a triangle with angles 120°, 50°, 10°. 5+5=10

3 a)Represent the standard fuzzy set operations using Venn diagram.

b) Write down the difference between crisp set and fuzzy set. 5+5=10

4.(a)Write and explain examples of fuzzy proposition.
(b) Consider some fuzzy sets and prove the distributive law, associative law.
(c) Draw and explain models of Neuro Fuzzy System. 3+3+4=10

5. The membership function distribution of a fuzzy set is assumed to follow a Gaussian distribution with mean m=100 and standard deviation σ=20. Determine 0.6-cut of this distribution. 10

6. Explain any five defuzzification methods. 10

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